

PEBL SESSION PREPARATION

Example of how deeply we'll learn

Suppose they ask:

What is the `<html>` tag?

Most beginners answer:

It is the root tag.

That is correct, but an interviewer may ask:

Why is it called the root tag?

A better answer is:

"The `<html>` tag is called the root element because every visible and invisible element in an HTML document must be placed inside it. It acts as the parent of the `<head>` and `<body>` sections."

Then they may ask:

Can we have two `<html>` tags?

Answer:

✗ No.

An HTML document can have only one root `<html>` element.

Then they may ask:

Which are the children of `<html>`?

Answer:

`<html>`

`<head>`

`</head>`

`<body>`

`</body>`

`</html>`

The children are:

- `<head>`
 - `<body>`
-

Then they may ask:

Which is the parent of `<title>`?

Answer:

`<head>`

Then:

Which is the grandparent of `<title>`?

Answer:

`<html>`

Then:

Can we write `<body>` before `<head>`?

Technically browsers may still try to render the page, but according to the standard HTML structure, `<head>` should come before `<body>`.

See?

This is how interviewers go deeper.

Another Example

They ask:

"What is `<head>`?"

Most students answer:

It stores metadata.

The interviewer asks:

What is metadata?

Good answer:

"Metadata is information about the webpage that is mainly used by the browser and search engines. It is generally not displayed directly on the webpage."

"The <head> tag is a container that stores information about the webpage, known as metadata. This information is not displayed on the webpage itself but is used by the browser and search engines. It can contain elements like <title>, <meta>, <link>, <style>, and <script>."

Then they ask:

Can users see the <title> tag?

Answer:

No. They don't see it inside the webpage. It appears on the browser tab.

Then:

Can users see the <meta> tag?

No.

It provides information like character encoding, viewport settings, description, keywords, etc., but it isn't displayed as page content.

Topic 1: What is HTML?

Interview Question:

What is HTML?

Basic Answer

HTML stands for HyperText Markup Language. It is the standard markup language used to create the structure of web pages.

Professional Answer (Best for Interview)

HTML (HyperText Markup Language) is the standard markup language used to structure the content of a webpage. It tells the browser what elements such as headings, paragraphs, images, links, tables, and forms should appear on the webpage. HTML creates the structure, while CSS is used for styling and JavaScript adds functionality.

Tricky Question

Is HTML a programming language?

Answer:

No.

Why?

A programming language can make decisions, perform calculations, and use loops and conditions.

Examples:

- **Java**
- **Python**
- **JavaScript**
- **C++**

HTML cannot do these things.

It only describes the structure of a webpage.

So HTML is a Markup Language, not a programming language.

What is Markup?

Suppose you write:

Welcome to ChatGPT

How does the browser know it's a heading?

It doesn't.

So we "mark" it using tags.

<h1>Welcome to ChatGPT</h1>

The <h1> tag tells the browser:

"This is a heading."

This process is called marking up the content.

That's why it's called a Markup Language.

"Markup means adding tags to content to define its structure and meaning. HTML uses markup tags to tell the browser what each piece of content represents. For example, if we write plain text, the browser cannot know which text is a heading and which is a paragraph. By using tags like <h1> for a heading and <p> for a paragraph, we structure the content so that the browser, developers, and users can understand it correctly."

What is HyperText?

This is one of the most frequently asked interview questions.

Many students know HTML stands for HyperText Markup Language, but they don't know what HyperText means.

HyperText means text that contains links to another page or document.

Example:

```
<a href="about.html">About Us</a>
```

When you click it, it opens another page.

This connected text is called HyperText.

What is a Tag?

Interview Question:

What is a tag?

Answer:

A tag is a keyword enclosed within angle brackets (< >) that tells the browser how to display or structure content.

Example:

```
<h1>Hello</h1>
```

Here,

- <h1> is the opening tag.
 - </h1> is the closing tag.
 - Hello is the content.
-

What is an Element?

Many students confuse tag and element.

Tag

<h1>

This is only a tag.

Element

<h1>Hello</h1>

This complete structure (opening tag + content + closing tag) is called an element.

Interview Question

Difference between Tag and Element?

Answer:

Tag

- Starts with < >
- Gives instructions to the browser
- Can be opening or closing

Element

- Complete structure
 - Opening tag + content + closing tag
-

What is an Attribute?

Example:

Here:

src and alt are attributes.

They provide additional information about an element.

Tricky Question

Where are attributes written?

Answer:

Inside the opening tag.

Example:

href is inside the opening tag.

HTML File Extension

Question:

What is the extension of an HTML file?

Answer:

.html

Examples:

index.html

home.html

about.html

Why is the first page usually named index.html?

When someone visits a website like:

www.abc.com

the web server automatically looks for a default page, which is commonly named index.html.

That's why the home page is often named index.html.

HTML Document Structure (Boilerplate)

Every HTML page starts with:

<!DOCTYPE html>

<html lang="en">

```
<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>My Website</title>

</head>


<body>


</body>


</html>
```

An interviewer may ask you to explain this line by line.

Let's do that.

Why do we use `<meta charset="UTF-8">`?

Answer:

We use `<meta charset="UTF-8">` to tell the browser that the webpage uses UTF-8 character encoding, so text and special characters from different languages are displayed correctly.

We use the viewport meta tag to make webpages responsive so they display correctly on different screen sizes such as mobiles, tablets, and desktops.

1. `<!DOCTYPE html>`

Question:

What is `<!DOCTYPE html>`?

Answer:

It tells the browser that the document is written in HTML5. This helps the browser render the page using modern HTML standards.

Tricky Question:

Is <!DOCTYPE html> a tag?

Answer:

No. It is a declaration, not an HTML tag.

2. <html>

This is the root element of the HTML document.

Everything except the <!DOCTYPE html> declaration is placed inside it.

Example:

<html>

<head>

</head>

<body>

</body>

</html>

Children of <html>

- **<head>**
- **<body>**

Parent of <head>?

<html>

Parent of <body>?

<html>

3. lang="en"

What is it?

It is an attribute of the <html> tag.

It specifies the language of the webpage.

Example:

<html lang="en">

English

<html lang="ta">

Tamil

<html lang="hi">

Hindi

It helps search engines, browsers, and screen readers understand the page language.

Topic 2: The <head> Tag

Interview Question:

What is the <head> tag?

Beginner Answer

The <head> tag contains metadata about the webpage.

Correct, but not enough.

Professional Interview Answer

The <head> section contains information about the webpage that is mainly used by the browser, search engines, and other web services. The content inside the <head> is generally not displayed on the webpage itself.

Example:

<head>

<title>Amazon</title>

</head>

The browser shows Amazon on the browser tab, not inside the webpage.

Tricky Question

Can we see the contents of the <head> tag?

Answer:

Mostly No.

The information inside <head> is meant for the browser, not for users.

However:

The title appears on the browser tab.

Everything else is generally hidden from the webpage.

Parent and Child Relationship

<html>

<head>

</head>

<body>

</body>

</html>

Parent of <head>?

Answer:

<html>

Children inside <head>?

Usually

- **title**
- **meta**
- **link**
- **style**
- **script**
- **base**

Interview Question

Can we write an `<h1>` tag inside `<head>`?

Answer:

✗ No.

Why?

Because `<head>` is not meant to display webpage content.

`<h1>` is visible content.

Visible content belongs inside the `<body>`.

`<title>` Tag

Example

`<title>My Website</title>`

What is the purpose of the `<title>` tag?

Answer:

The `<title>` tag specifies the title of the webpage.

It appears:

- Browser tab
 - Search engine results
 - Bookmarks
 - Browser history
-

Tricky Question

How many `<title>` tags should a webpage have?

Answer:

Only one.

Can we put `<title>` inside `<body>`?

Answer:

No.

It belongs inside <head>.

<meta> Tag

This is where interviewers start asking tricky questions.

What is the <meta> tag?

Answer:

The <meta> tag provides additional information (metadata) about the webpage.

It is not displayed on the webpage.

What is Metadata?

Interviewers LOVE asking this.

Best Answer:

Metadata means "data about data." It provides information about the webpage rather than displaying actual webpage content.

Examples

- **character encoding**
 - **viewport settings**
 - **author**
 - **description**
 - **keywords**
-

Meta Charset

<meta charset="UTF-8">

Why do we use UTF-8?

Suppose your webpage contains:

தமிழ்

日本



Without UTF-8

Browser may display

@#\$\$%^&

?????

UTF-8 allows the browser to correctly display characters from many languages and symbols.

Interview Question

What happens if we remove UTF-8?

Answer:

Some special characters may display incorrectly.

Viewport Meta Tag

<meta

name="viewport"

content="width=device-width, initial-scale=1.0">

Students usually memorize this.

Interviewers ask:

Why?

Why do we use viewport?

Imagine your website on:

Desktop

1920 pixels

Phone

390 pixels

Without the viewport meta tag, the browser may render the page as if it's a desktop page, making everything tiny on a phone.

The viewport tag tells the browser to match the page width to the device width.

Meaning of

width=device-width

It tells the browser:

"Use the width of the current device."

Meaning of

initial-scale=1.0

Display the webpage at its normal zoom level when it first loads.

Interview Question

Is viewport useful for responsive websites?

Answer:

Yes.

Without it, responsive CSS may not behave as expected on mobile devices.

Meta Description

Example

<meta

name="description"

content="Online shopping website">

Purpose?

Provides a short description of the webpage.

Search engines may use this description in search results.

Meta Keywords

Example

```
<meta  
name="keywords"  
content="HTML,CSS,JavaScript">
```

Years ago, search engines used this more heavily.

Today, major search engines like Google don't rely on it for ranking, but you may still see it in older code or some projects.

Meta Author

```
<meta  
name="author"  
content="Ashvi">
```

Purpose?

Specifies who created the webpage.

Quick Interview

What is the difference?

title

vs

meta description

Answer

Title

Appears on the browser tab.

Meta description

Provides information about the webpage and may appear in search engine results.

<link> Tag

Example

```
<link rel="stylesheet" href="style.css">
```

Question

Why do we use <link>?

Answer

It connects external resources to the HTML page.

Usually

CSS

Question

Can CSS work without <link>?

Answer

Yes.

There are three ways:

- **Inline CSS**
- **Internal CSS**
- **External CSS**

The <link> tag is used for external CSS.

<script> Tag

Example

```
<script src="script.js"></script>
```

Purpose?

Connects JavaScript to the webpage.

Interview Question

Can JavaScript be written without an external file?

Answer

Yes.

Inside

<script>

alert("Hello")

</script>

Difference

<link>

Connects CSS or other external resources.

<script>

Connects or contains JavaScript.

Very Tricky Questions

1. Can a webpage run without a <head> tag?

Browsers may try to create one automatically, but in a properly structured HTML document, you should include a <head> section.

2. Can there be two <head> tags?

✗ No.

Only one <head> section should exist.

3. Can there be two <body> tags?

✗ No.

Only one <body> element should exist.

4. Which loads first?

<head> or <body>?

Answer:

The browser processes the <head> first because it contains important information like the page title, CSS links, and metadata before rendering the page content.

5. Where is the visible content of the webpage written?

Inside

<body>

PART 3 – The <body> Tag (Deep Interview Preparation)

Question 1: What is the <body> tag?

Beginner Answer

The <body> tag contains the visible content of the webpage.

Correct.

Professional Interview Answer

The <body> tag contains all the content that is displayed to the user in the browser, such as headings, paragraphs, images, links, tables, forms, videos, and buttons.

Example:

<body>

<h1>Welcome</h1>

<p>Hello Everyone</p>

</body>

Everything inside <body> is displayed on the webpage.

Tricky Question

Can we have two <body> tags?

✗ No.

Every HTML document should have only one <body> element.

Another Tricky Question

What happens if we write content outside <body>?

Example:

```
<h1>Hello</h1>
```

```
<body>
```

```
</body>
```

Answer:

Browsers may try to fix the HTML automatically, but this is not the correct HTML structure. All visible content should be inside the <body> element.

Parent and Child Relationship

```
<html>
```

```
<head>
```

```
</head>
```

```
<body>
```

```
<h1>Hello</h1>
```

```
<p>Welcome</p>
```

```
</body>
```

```
</html>
```

Interviewer:

Parent of <body>?

Answer:

<html>

Children of <body>?

- **h1**
- **p**
- **img**
- **div**
- **section**
- **article**
- **table**
- **form**

Almost every visible HTML element can be inside <body>.

Heading Tags

One of the favorite interview questions.

HTML provides

h1

h2

h3

h4

h5

h6

Why do we have six headings?

Think of a book.

Book Name

Chapter

Topic

Subtopic

Point

Subpoint

Websites also need this hierarchy.

Example

<h1>HTML Tutorial</h1>

<h2>Introduction</h2>

<h3>History</h3>

<h4>HTML5</h4>

<h5>Features</h5>

<h6>Summary</h6>

Interview Question

Which heading is biggest?

Answer

<h1>

Which is smallest?

<h6>

Very Tricky Question

Can we use

<h1>

ten times?

Most beginners say:

"No."

That answer is not completely correct.

Correct Answer

HTML allows multiple <h1> elements.

However, for a well-structured page, it's generally recommended to have one main <h1> representing the primary topic, with <h2>, <h3>, etc., used for subsections.

Another Tricky Question

Does

<h1>

make text important?

Answer

No.

It indicates the highest-level heading in the document structure.

If you only want text to look bigger, CSS should be used for styling.

Paragraph Tag

<p>

Hello

`</p>`

Purpose

Represents a paragraph of text.

Interview Question

Can paragraphs contain multiple sentences?

Yes.

A paragraph can contain one sentence or many sentences.

Tricky Question

Can a paragraph contain another paragraph?

`<p>`

`<p>Hello</p>`

`</p>`

Answer

✗ No.

A `<p>` element cannot contain another `<p>` element.

Line Break

`<br>`

Purpose

Moves the next content to a new line.

Example

Hello

World

**Without
**

Output

Hello World

**With
**

**Hello
World**

Output

Hello

World

Interview Question

**Does
 have a closing tag?**

Answer

No.

It is a void (empty) element.

Horizontal Rule

<hr>

Purpose

Creates a horizontal line to separate sections.

Interview Question

Does <hr> need a closing tag?

No.

It is also a void element.

<pre> Tag

Very favorite interview question.

Example

<pre>

Hello

World

</pre>

Output

Hello

World

Without <pre>

Browser removes extra spaces.

Example

Hello World

Output

Hello World

With <pre>

Everything is preserved.

Interview Question

Difference between

<p>

and

<pre>

<p>

<pre>

Ignores extra spaces

Preserves spaces

Ignores extra line breaks

Preserves line breaks

Normal paragraph

Preformatted text

Comments

Example

<!-- This is a comment -->

Purpose

Comments are for developers.

They are ignored by the browser when rendering the page.

Interview Question

Can users normally see comments?

No.

They are not displayed on the webpage.

Whitespace

Interviewers love this.

Example

Hello

World

Browser shows

Hello World

Why?

HTML collapses multiple spaces and line breaks into a single space (except inside elements like <pre>).

Nested Elements

Example

<div>

<h1>Hello</h1>

<p>Welcome</p>

</div>

Here

Parent

div

Children

h1

p

Interview Question

What is nesting?

Answer

Placing one HTML element inside another HTML element.

Empty (Void) Elements

These elements don't have closing tags.

Examples

**
**

<hr>

<input>

<meta>

<link>

Interview Question

Why don't they have closing tags?

Because they don't wrap content. They perform a single purpose and don't contain inner text or child elements.

PART 4 – Formatting Tags (Deep Interview Preparation)

What are Formatting Tags?

Interview Question:

What are formatting tags?

Professional Answer:

Formatting tags are HTML tags used to format or emphasize text. They help make text bold, italic, underlined, highlighted, smaller, deleted, inserted, subscript, or superscript.

Example:

<p>I am learning HTML.</p>

Output:

I am learning HTML.

1. Tag

Example:

HTML

Output:

HTML

Purpose

Makes text bold.

Interview Question

What does the `<b>` tag do?

Answer:

It simply makes the text bold without giving it any special meaning.

2. `<strong>` Tag

Example:

`<strong>Warning!</strong>`

Output:

Warning!

Looks the same as `<b>`.

So why do we have two tags?

★ **Very Important Interview Question**

Difference between `<b>` and `<strong>`

This is one of the most frequently asked HTML questions.

<code></code>	<code></code>
Makes text bold	Makes text bold
No special meaning	Indicates the text is important
Mainly visual	Has semantic meaning

Professional Answer

The `<b>` tag is used only to make text bold for visual presentation. The `<strong>` tag also usually appears bold, but it tells browsers and assistive technologies that the text is important.

Real Example

`<p>Please read the <b>manual</b>.</p>`

Only bold.

`<p><strong>Warning:</strong> High Voltage.</p>`

Here, "Warning" is important.

Interview Trick

Which one is better?

Answer:

`<strong>`

Because it provides meaning, not just appearance.

3. `<i>` Tag

Example

`<i>HTML</i>`

Output:

HTML

Purpose:

Makes text italic.

4. `<em>` Tag

Example

`<em>Important</em>`

Output:

Important

Looks the same.

★ Interview Question

Difference between `<i>` and `<em>`?

`<i>`

`<em>`

Italic text

Emphasized text

Visual formatting Semantic meaning

Professional Answer

The `<i>` tag is used only for italic styling. The `<em>` tag emphasizes text and adds meaning, which can be understood by browsers and screen readers.

Example

`<p>I like Harry Potter.``</p>`

Book title.

`<p>You must submit today.``</p>`

Emphasizes the word "must".

5. `<u>` Tag

Example

`<u>HTML``</u>`

Output

`<u>HTML``</u>`

Purpose

Underlines text.

Interview Question

Should we underline ordinary text?

Answer:

Usually no, because underlined text can look like a hyperlink.

6. `<mark>` Tag

Example

`<mark>HTML</mark>`

Output

Highlighted text.

Purpose

Highlights important text, similar to using a highlighter pen.

Real Example

`<p>Exam topic: <mark>Forms</mark></p>`

7. `<small>` Tag

Example

`<small>Terms and Conditions</small>`

Output

Smaller text.

Purpose

Displays text in a smaller size, often used for notes or legal information.

8. `<del>` Tag

Example

`<del>$100</del>`

Output

~~\$100~~

Purpose

Shows deleted or removed text.

Example

Old Price

\$100

New Price

\$80

9. <ins> Tag

Example

<ins>\$80</ins>

Output

<u>\$80</u>

Purpose

Indicates newly inserted text.

Difference

<u>

Just underlines text.

<ins>

Represents newly inserted content.

10. <sub> Tag

Subscript

Example

H₂O

Output

H₂O

Purpose

Displays text slightly below the normal line.

Real Uses

- **Chemical formulas**
 - **Mathematical expressions**
-

11. <sup> Tag

Superscript

Example

2³

Output

2³

Purpose

Displays text slightly above the normal line.

Real Uses

- **Exponents**
 - **Footnotes**
-

★ Interview Trick

Difference between

<sub>

and

<sup>

<sub> <sup>

Below the line Above the line

H₂O x²

Nesting Formatting Tags

Example

```
<strong><em>HTML</em></strong>
```

Output

HTML

Interview Question

Can formatting tags be nested?

Answer:

Yes.

HTML elements can be nested as long as they are properly opened and closed.

Correct

```
<strong>
```

```
<em>
```

HTML

```
</em>
```

```
</strong>
```

Wrong

```
<strong>
```

```
<em>
```

HTML

`</strong>`

`</em>`

Tags must close in the reverse order in which they were opened.

★ Super Tricky Questions

1. Does `<strong>` always appear bold?

Usually yes, but browsers can style it differently.

The important point is its meaning, not just its appearance.

2. Can CSS make text bold instead of `<b>`?

Yes.

Example

`font-weight: bold;`

3. Can CSS make text italic?

Yes.

`font-style: italic;`

4. If CSS can do everything, why use formatting tags?

Excellent interview question.

Answer:

CSS controls how content looks.

HTML describes what the content means.

For example:

`<strong>Danger</strong>`

This tells browsers and assistive technologies that the word is important.

★ Real Interview Conversation

Interviewer:

What is the difference between HTML and CSS?

You:

HTML provides the structure and meaning of the content, while CSS controls the presentation and appearance of that content.

Interviewer:

If CSS can make text bold, why use ?

You:

Because conveys that the content is important, whereas CSS only changes its visual appearance. HTML provides semantics, CSS provides styling.

PART 5 – Lists in HTML (Deep Interview Preparation)

What is a List?

Interview Question

What is a list in HTML?

Professional Answer

A list is used to display related items in an organized manner. HTML provides ordered lists, unordered lists, and description lists.

Example:

Shopping List

- **Milk**
- **Bread**
- **Eggs**

Instead of writing

Milk Bread Eggs

we organize them as a list.

Types of Lists

There are 3 types.

1. Unordered List ()
 2. Ordered List ()
 3. Description List (<dl>)
-

1. Unordered List ()

Example

```
<ul>
```

```
  <li>HTML</li>
```

```
  <li>CSS</li>
```

```
  <li>JavaScript</li>
```

```
</ul>
```

Output

- HTML
 - CSS
 - JavaScript
-

Interview Question

Why is it called an unordered list?

Answer:

Because the items are not arranged in a specific sequence. The order doesn't matter.

Example:

Favorite Fruits

- Apple
- Mango
- Orange

Whether Apple comes first or Mango comes first doesn't change the meaning.

Parent-Child Relationship

`<ul>`

`<li>HTML</li>`

`<li>CSS</li>`

`</ul>`

Parent

`<ul>`

Children

`<li>`

Tricky Question

Can we write text directly inside `<ul>`?

Wrong

`<ul>`

HTML

`<li>CSS</li>`

`</ul>`

Correct

`<ul>`

`<li>HTML</li>`

`<li>CSS</li>`

A should contain elements, not plain text.

2. Ordered List ()

Example

Wake up

Brush

Study

Output

- 1. Wake up**
 - 2. Brush**
 - 3. Study**
-

Interview Question

Why do we use an ordered list?

Answer

When the sequence matters.

Example

Recipe

- 1. Boil water**
- 2. Add tea powder**
- 3. Add milk**

Changing the order changes the result.

**Difference Between and **

Unordered

Ordered

Bullet points

Numbers or letters

Sequence not important

Sequence important

3. List Item ()

Question

What is ?

Answer

It represents a single item inside a list.

Example

Apple

Mango

Orange

There are three list items.

Interview Question

Can exist without or ?

Technically, browsers may try to display it, but according to HTML semantics, `<li>` should be placed inside a list container like `<ul>`, `<ol>`, or (rarely) `<menu>`.

Ordered List Types

Many interviewers ask this.

Example

```
<ol type="A">
```

```
<li>HTML</li>
```

```
<li>CSS</li>
```

```
</ol>
```

Output

A. HTML

B. CSS

Possible Values

1

A

a

I

i

Example

```
<ol type="I">
```

```
<li>HTML</li>
```

`<li>CSS</li>`

`</ol>`

Output

I. HTML

II. CSS

Interview Question

Default type of ?

Answer

Numbers

1

2

3

Start Attribute

Example

`<ol start="5">`

`<li>HTML</li>`

`<li>CSS</li>`

`</ol>`

Output

5. HTML

6. CSS

Interview Question

Why do we use start?

Answer

To begin numbering from a specific value.

Reversed Attribute

Example

<ol reversed>

HTML

CSS

JavaScript

Output

3. HTML

4. CSS

5. JavaScript

Interview Question

Purpose of reversed?

Displays the list in reverse numbering.

Description List (<dl>)

Many beginners forget this, but interviewers may ask it.

Example

<dl>

<dt>HTML</dt>

<dd>HyperText Markup Language</dd>

<dt>CSS</dt>

<dd>Cascading Style Sheets</dd>

</dl>

What is <dl>?

Description List

What is <dt>?

Description Term

Example

HTML

What is <dd>?

Description Details

Example

HyperText Markup Language

Parent Relationship

<dl>

<dt>

<dd>

</dl>

Parent

<dl>

Children

<dt>

<dd>

Nested Lists

Very common interview question.

Example

Programming

HTML

CSS

Output

- **Programming**

- HTML
- CSS

Interview Question

What is a nested list?

Answer

A list placed inside another list.

Tricky Questions

1. Can we put a inside an ?

 **Yes.**

Example

Frontend

HTML

CSS

2. Can we put an inside a ?

✓ Yes.

3. Can a contain another directly?

Instead of:

```
<ul>
  <ul>
    <li>HTML</li>
  </ul>
</ul>
```

the correct approach is to nest the second list inside an :

```
<ul>
  <li>
    Frontend
    <ul>
      <li>HTML</li>
    </ul>
  </li>
</ul>
```

4. Which tag creates bullets?

Answer

```
<ul>
```

5. Which tag creates numbering?

Answer

```
<ol>
```

6. Which tag represents each item?

Answer

Real Interview

Interviewer

What are the different types of lists?

You

HTML provides three types of lists:

- **Unordered List ()**
 - **Ordered List ()**
 - **Description List (<dl>)**
-

Interviewer

When would you use instead of ?

You

I use when the order of items is important, such as instructions, recipes, or steps in a process. I use when the order doesn't matter, such as a list of hobbies or technologies.

PART 6 – Links (<a> Tag) – Deep Interview Preparation

What is a Hyperlink?

Interview Question

What is a hyperlink?

Professional Answer

A hyperlink is a clickable element that allows users to navigate from one webpage, file, email address, phone number, or section of a webpage to another.

Example:

Click Here

When clicked, it opens another page.

What is the <a> Tag?

Interview Question

What is the <a> tag?

Professional Answer

The <a> (Anchor) tag is used to create hyperlinks in HTML.

Example

```
<a href="https://www.google.com">Google</a>
```

Output

Google (clickable)

Why is it called Anchor?

Many students don't know this.

Originally, an anchor means something that marks or connects a location.

The <a> tag connects one location to another.

That's why it is called the Anchor tag.

Basic Syntax

```
<a href="URL">Link Text</a>
```

Three parts

<a

Opening tag

href=""

Destination

Link Text

Visible clickable text

What is href?

Interviewers ask this almost every time.

Answer

href stands for

Hypertext Reference

It specifies the destination of the hyperlink.

Example

```
<a href="about.html">About</a>
```

When clicked

Browser opens

about.html

Tricky Question

Can an <a> tag work without href?

Example

```
<a>Google</a>
```

Answer

Yes, it is still an anchor element, but without href it does not function as a normal hyperlink. It won't navigate anywhere when clicked.

Absolute URL

Example

```
<a href="https://www.google.com">
```

Google

```
</a>
```

Question

What is an absolute URL?

Answer

A complete web address including the protocol (https://) and the domain name.

Example

https://www.google.com

Relative URL

Example

About

Question

What is a relative URL?

Answer

A path that points to another file within the same website or project.

Interview Question

Difference between Absolute URL and Relative URL?

Absolute URL

Relative URL

Complete web address

Local file path

Includes https://

Doesn't include domain

Can open external websites

Used within the same project

Example

Project

Project

index.html

about.html

contact.html

Inside

index.html

You can write

```
<a href="about.html">
```

About

```
</a>
```

No need to write

<https://mywebsite.com/about.html>

Opening Links in New Tab

Example

```
<a
```

```
href="https://www.google.com"
```

```
target="_blank">
```

Google

```
</a>
```

Question

What does

```
target="_blank"
```

mean?

Answer

Opens the linked page in a new browser tab (or window, depending on browser settings).

What happens if we don't use it?

Browser opens the link in the same tab by default.

Interview Question

Default value of target?

Answer

`_self`

Meaning

Open in the same tab.

Email Link

Example

`<a href="mailto:abc@gmail.com">`

Email Us

`</a>`

Purpose

Opens the user's default email application to compose a new email.

Telephone Link

Example

`<a href="tel:+911234567890">`

Call Us

`</a>`

Purpose

Allows users to call directly, especially useful on mobile devices.

Download Attribute

Example

`<a`

`href="resume.pdf"`

`download>`

Download Resume

Purpose

Suggests that the browser download the linked file instead of opening it.

Image as Link

Interviewers love practical questions.

Example

```
<a href="index.html">  
    
</a>
```

Question

Can an image become a link?

Answer

Yes.

Wrapping an inside an <a> makes the image clickable.

Linking to a Section (Bookmark)

Suppose

```
<h2 id="contact">
```

Contact Us

```
</h2>
```

Now

```
<a href="#contact">
```

Go to Contact

```
</a>
```

Question

What happens?

Browser scrolls to the element whose id is "contact".

Linking Between Pages

Example

Home Page

`<a href="about.html">`

About

`</a>`

Inside About Page

`<a href="index.html">`

Home

`</a>`

Nesting

Can we write

`<a>`

`<strong>`

Google

`</strong>`

`</a>`

Yes.

Perfectly valid.

Can We Put a Link Inside Another Link?

Wrong

```
<a href="page1.html">
```

```
<a href="page2.html">
```

Click

```
</a>
```

```
</a>
```

Answer

✗ No.

Anchor elements cannot be nested inside another anchor element.

Anchor Around a Button

Example

```
<a href="about.html">
```

```
<button>
```

About

```
</button>
```

```
</a>
```

This works in browsers, but it's generally better practice to style an <a> to look like a button when the action is navigation, and use a <button> for actions like submitting a form.

Interview Trick

Difference between Button and Link?

Button

Performs an action

Examples

- **Submit form**
 - **Delete record**
 - **Save data**
-

Link

Navigates

Examples

- **Home**
 - **Contact**
 - **About**
-

Common Mistakes

Mistake 1

<a>

Google

No href

Not useful as a hyperlink.

Mistake 2

Google

`</a>`

Empty destination.

Usually reloads or stays on the current page.

Mistake 3

`<a href="google.com">`

Google

`</a>`

For external websites, it's better to include the full URL, such as:

`<a href="https://www.google.com">`

Google

`</a>`

Real Interview

Interviewer

What is the difference between

href

and

src

Excellent question.

href

src

Refers to another resource Embeds a resource

Used in `<a>` and `<link>` Used in `<img>`, `<script>`, etc.

Example

`<a href="about.html">`

Navigates

Example

```

```

Displays the image.

Super Tricky Questions

1. Can an <a> tag contain an image?

✓ Yes.

2. Can it contain a paragraph?

✓ Yes.

In HTML5, an anchor can wrap many kinds of content, as long as you don't nest another <a> inside it.

3. Can we style links using CSS?

✓ Yes.

4. Is href mandatory?

For a functioning hyperlink, yes. Without it, the anchor won't navigate.

PART 7 – Images (Tag) – Deep Interview Preparation

Interview Question

What is the tag?

Professional Answer

The tag is used to display images on a webpage.

Example:

```

```

Output:

An image of a cat appears on the webpage.

Is a Container Tag?

Answer:

✗ No.

It is a Void (Empty) Element.

It has:

- Opening tag 
- Closing tag 

Correct

Wrong

Why doesn't have a closing tag?

Excellent interview question.

Because it does not contain any content inside it.

Unlike

<p>Hello</p>

The tag only tells the browser:

"Display this image."

It doesn't wrap text or child elements.

What is src?

One of the most asked questions.

Example

src stands for

Source

It tells the browser where the image is located.

Tricky Question

What happens if src is wrong?

Example

```

```

But

dog.jpg

doesn't exist.

Output

Image won't be displayed.

Instead, the browser may show the broken image icon, and if alt is provided, it may display the alternative text.

What is alt?

Another favorite interview question.

Example

```

```

What is Alt?

Professional Answer

The alt attribute provides alternative text for an image if the image cannot be displayed. It also helps screen readers describe the image to visually impaired users.

Interview Trick

Does alt display when the image loads successfully?

Answer

Usually No.

If the image loads correctly,

alt

remains hidden.

It appears (or is used by assistive technologies) only when needed.

Why is alt Important?

Suppose

```

```

But

lion.jpg

is deleted.

Instead of

✖ Blank space

Browser can display

Lion

This helps users understand what was supposed to be there.

Accessibility

Interviewers may ask:

Why is alt important for accessibility?

Answer:

People using screen readers cannot see images.

The screen reader reads the alt text aloud, helping them understand the image.

Width Attribute

Example

```

```

Purpose

Sets the image width.

Height Attribute

Example

```

```

Purpose

Sets the image height.

Interview Question

Can we use both width and height?

Yes.

Example

```

```

Tricky Question

What happens if the width and height don't match the image's original aspect ratio?

Example

Original

300×300

You set

300×100

The image may appear stretched or squashed.

Better Practice

Usually, specify only one dimension (or use CSS) so the browser preserves the image's aspect ratio automatically.

Relative Image Path

Suppose

Project

index.html

images

cat.jpg

Code

<img

src="images/cat.jpg">

This is called a

Relative Path

Absolute Path

Example

<https://website.com/images/cat.jpg>

Complete web address.

Difference

Relative Path

images/cat.jpg

Absolute Path

<https://abc.com/images/cat.jpg>

Interview Question

Which one is commonly used inside projects?

Answer

Relative paths.

Image Formats

Interviewers sometimes ask this.

Common formats

- **JPG / JPEG**
 - **PNG**
 - **GIF**
 - **SVG**
 - **WebP**
-

JPG

Best for

Photographs

Advantages

- **Smaller file size**
- **Good for photos**

Disadvantage

- **Doesn't support transparency**
-

PNG

Best for

Logos

Icons

Advantages

Supports transparency.

GIF

Best for

Simple animations.

SVG

Very important.

Best for

- **Logos**
- **Icons**
- **Shapes**

Advantages

Can scale without losing quality.

WebP

Modern format.

Advantages

- **Smaller file size**
- **Good quality**
- **Supports transparency**

Many modern websites use WebP for better performance.

Interview Question

Which format is best for company logos?

Best Answer

SVG (if available) because it scales cleanly. PNG is also commonly used, especially when an SVG version isn't available.

Which format is best for photographs?

JPG or WebP.

Lazy Loading

Very good interview topic.

Example

```

```

Purpose

Delays loading the image until it's close to entering the user's viewport.

Question

Why?

Answer

Improves page loading speed, especially on pages with many images.

Figure and Figcaption

Many freshers don't know this.

Example

```
<figure>  
  
  
  
<figcaption>
```

Sleeping Cat

</figcaption>

</figure>

Purpose

<figure>

Groups an image (or other media) with its caption.

<figcaption>

Provides a caption describing the figure.

Common Mistakes

Mistake 1

Wrong file name

Actual file

cat.jpg

On many web servers, filenames are case-sensitive, so this can fail.

Mistake 2

Wrong folder

Actual

images/cat.jpg

Image won't load.

Mistake 3

Missing file extension

Wrong

Correct

Mistake 4

No alt

Better

Real Interview

Interviewer

Difference between

src

and

alt

Answer

src	alt
------------	------------

Specifies the image location	Provides alternative text
-------------------------------------	----------------------------------

Required to load the image	Helps if the image can't be displayed and improves accessibility
-----------------------------------	---

Interviewer

Can an image become a hyperlink?

Answer

Yes.

Example

```
<a href="index.html">
```

```

```

```
</a>
```

Interviewer

Can we display an image without alt?

Answer

Yes, the image can still display.

However, providing meaningful alt text is recommended for accessibility and for situations where the image cannot be loaded.

Super Tricky Questions

1. Is alt compulsory?

Technically, an element can be written without alt, but good HTML practice and accessibility guidelines recommend including it. Decorative images may intentionally use alt="".

2. Does alt improve SEO?

Yes.

It helps search engines better understand what the image represents.

3. Can we display images from another website?

Yes.

Example

```
<img
```


src="https://example.com/image.jpg"

alt="Example">

provided you have permission and the remote server allows it.

PART 8 – HTML Tables (Complete Interview Preparation)

Question 1

What is a Table?

Professional Answer

An HTML table is used to display data in rows and columns.

Example:

Name Age City

Ashvi 22 Chennai

Rahul 21 Bangalore

Basic Structure

<table>

<tr>

<th>Name</th>

<th>Age</th>

</tr>

<tr>

<td>Ashvi</td>

<td>22</td>

</tr>

</table>

Parent-Child Relationship

This is one of the favorite interview topics.

table

|

|— tr

| |— th

| |— th

|

|— tr

|— td

|— td

Question

Parent of <tr>?

Answer

<table>

Parent of <td>?

Answer

<tr>

Parent of <th>?

Answer

<tr>

<table> Tag

Question

What is the purpose of the <table> tag?

Professional Answer

The <table> element is the container that holds all table-related elements such as rows, cells, headers, and captions.

<tr> Tag

Question

What is <tr>?

Answer

tr

stands for

Table Row

Each <tr> creates one row.

Example

<tr>

<td>HTML</td>

<td>CSS</td>

</tr>

One row.

<td> Tag

Question

What is <td>?

Answer

td

stands for

Table Data

It represents a normal data cell.

Example

<td>Chennai</td>

<th> Tag

Question

What is <th>?

Answer

th

stands for

Table Header

Example

<th>Name</th>

Difference Between <th> and <td>

<th>

<td>

Header Cell

Data Cell

Usually bold by default

Normal text by default

Usually centered by default

Usually left-aligned by default

Interview Trick

Does <th> automatically become bold?

Yes, browsers typically render it bold by default.

But remember:

That is browser styling.

Its real purpose is to identify a header cell, not simply to make text bold.

Example

```
<table>
```

```
<tr>
```

```
<th>Name</th>
```

```
<th>Age</th>
```

```
</tr>
```

```
<tr>
```

```
<td>Ashvi</td>
```

```
<td>22</td>
```

```
</tr>
```

```
</table>
```

Output

Name Age

Ashvi 22

```
<caption>
```

Many students forget this.

Example

<table>

<caption>

Student Details

</caption>

</table>

Output

Student Details

Purpose

Provides a title for the table.

Question

Where should <caption> be written?

Inside

<table>

Usually as the first child.

<thead>

Example

<thead>

<tr>

<th>Name</th>

<th>Age</th>

</tr>

</thead>

Purpose

Groups the header rows of a table.

<tbody>

Example

<tbody>

<tr>

<td>Ashvi</td>

<td>22</td>

</tr>

</tbody>

Purpose

Groups the main data rows.

<tfoot>

Example

<tfoot>

<tr>

<td>Total</td>

<td>2 Students</td>

</tr>

</tfoot>

Purpose

Groups footer rows, often used for totals or summaries.

Complete Table

<table>

<caption>

Student Details

</caption>

<thead>

<tr>

<th>Name</th>

<th>Age</th>

</tr>

</thead>

<tbody>

<tr>

<td>Ashvi</td>

<td>22</td>

</tr>

<tr>

<td>Rahul</td>

<td>21</td>

</tr>

</tbody>

<tfoot>

<tr>

`<td>Total</td>`

`<td>2</td>`

`</tr>`

`</tfoot>`

`</table>`

Interview Question

Is `<thead>` compulsory?

Answer

No.

The table works without it.

But it improves structure and readability.

`colspan`

One of the most important questions.

Example

`<th colspan="2">`

Student Details

`</th>`

Meaning

Merge

2 columns

Example

Before

| Name | Age |

After

| Student Details |

(one header spans both columns)

rowspan

Example

<td rowspan="2">

Ashvi

</td>

Meaning

Merge

2 rows

Difference

colspan

rowspan

Horizontal merge Vertical merge

Merges columns Merges rows

Interview Trick

Which one merges rows?

Answer

rowspan

Which one merges columns?

Answer

colspan

Nested Table

Can we place one table inside another?

Yes.

Example

<td>

<table>

...

</table>

</td>

Border

Old Method

<table border="1">

Today

We normally use CSS.

Example

table

{

border:1px solid black;

}

Cell Padding

Interview Question

What is Cell Padding?

Answer

Space

inside

the cell.

Example

| Hello |

More space around the word.

Cell Spacing

Space

between

cells.

Modern HTML uses CSS (padding, border-spacing) instead of old HTML attributes for this.

Common Mistakes

Wrong

```
<table>
```

```
<td>HTML</td>
```

```
</table>
```

Missing

```
<tr>
```

Correct

```
<table>
```

`<tr>`

`<td>HTML</td>`

`</tr>`

`</table>`

Wrong

`<tr>`

`<table>`

`</table>`

`</tr>`

A `<table>` should not be placed directly inside `<tr>`. If you need a nested table, put it inside a `<td>`.

Real Interview

Interviewer

Difference between

th

and

td

Best Answer

`<th>` defines a header cell.

`<td>` defines a normal data cell.

Interviewer

Difference between

thead

and

tbody

Answer

<thead>

contains header rows.

<tbody>

contains actual data rows.

Interviewer

Can a table work without <thead>?

Yes.

Interviewer

Can <td> exist without <tr>?

No.

A <td> must be inside a <tr>.

Super Tricky Questions

1. Which tag is the parent of every table element?

Answer

<table>

2. Can a table have multiple <tbody> elements?

 **Yes.**

A table can contain multiple <tbody> sections to logically group rows.

3. Can a table have multiple <thead> elements?

✗ No.

A table should have at most one <thead>.

4. Is <caption> compulsory?

No.

5. Can a table exist without <caption>?

Yes.

PART 9 – HTML Forms (Complete Interview Preparation)

Question 1

What is a Form?

Beginner Answer

A form is used to collect user information.

Correct.

Professional Interview Answer

An HTML form is used to collect data from users and send it to a server for processing.

Examples:

- **Login Form**
- **Registration Form**
- **Contact Form**
- **Feedback Form**
- **Search Box**

Whenever users enter data, they are interacting with a form.

Basic Structure

<form>

<input type="text">

<button>Submit</button>

</form>

Parent-Child Relationship

form

|

|— **label**

|— **input**

|— **textarea**

|— **select**

|— **button**

The <form> is the parent.

Question

What is the <form> tag?

Answer

The <form> tag acts as a container for form controls such as text fields, checkboxes, radio buttons, dropdowns, and buttons.

Can we write an <input> outside a <form>?

Example

<input type="text">

Yes.

The input will still appear.

But if you want to submit data together, the inputs should normally be placed inside a `<form>`.

The action Attribute

Example

```
<form action="/login">
```

What is action?

Professional Answer

The action attribute specifies where the submitted form data should be sent.

Example

```
<form action="success.html">
```

After submission

Browser sends data to

success.html

(or whatever server endpoint is configured).

Tricky Question

What happens if action is omitted?

Answer

The form submits to the current page by default.

The method Attribute

Example

```
<form  
action="/login"  
method="POST">
```

Question

What is method?

Answer

The method attribute specifies how the form data is sent.

There are two common methods:

- **GET**
 - **POST**
-

GET Method

Example

<form method="GET">

Suppose

Name

Ashvi

Browser URL becomes something like

example.com?name=Ashvi

Characteristics of GET

- **Data appears in the URL.**
 - **Suitable for searches and filters.**
 - **Not ideal for sensitive information like passwords.**
 - **Can be bookmarked.**
-

POST Method

Example

<form method="POST">

Characteristics

- **Data is sent in the request body.**
- **Doesn't appear in the URL.**
- **Better for passwords and personal information.**

- Commonly used for login and registration forms.

★ Interview Question

Difference Between GET and POST

GET	POST
Data in URL	Data in request body
Limited URL length	Can send much larger amounts of data
Can be bookmarked	Cannot be bookmarked in the same way
Good for searches	Good for login, registration, forms with sensitive data

Interview Trick

Which method is more secure?

Answer

POST is generally more suitable for sending sensitive data because it doesn't place the data in the URL.

⚠ However, HTTPS is what encrypts data during transmission. Using POST alone does not encrypt the data.

<label> Tag

Many freshers ignore this.

Interviewers don't.

Example

<label>Name</label>

<input type="text">

Purpose

Displays a label describing the input field.

Better Example

```
<label for="name">
```

Name

```
</label>
```

```
<input
```

```
id="name"
```

```
type="text">
```

What does for do?

The for attribute connects the label to an input.

The value of for

must match

the id

Example

```
<label for="email">
```

Email

```
</label>
```

```
<input
```

```
id="email">
```

Now clicking on

Email

also focuses the input.

Interview Question

Why use <label>?

Professional Answer

Labels improve usability and accessibility by clearly describing the associated form field and allowing users to focus the input by clicking the label.

<input> Tag

Question

What is <input>?

Answer

The <input> element is used to receive user input.

Is <input> a Void Element?

Yes.

<input>

No closing tag.

Input Types

This is where interviews become interesting.

There are many types.

1. Text

<input type="text">

Purpose

Single-line text input.

Example

Name

2. Password

<input type="password">

Purpose

Masks the entered characters.

Example

Interview Trick

Does it encrypt the password?

Answer

No.

It only hides the characters on the screen.

Encryption depends on how the data is transmitted (for example, over HTTPS) and handled by the server.

3. Email

<input type="email">

Purpose

Accepts email addresses.

Browsers can perform basic validation, such as checking for the presence of @.

4. Number

<input type="number">

Purpose

Accepts numeric values.

5. Date

<input type="date">

Displays a calendar/date picker.

6. Time

<input type="time">

Allows time selection.

7. Color

`<input type="color">`

Displays a color picker.

8. File

`<input type="file">`

Allows users to upload files.

9. Checkbox

`<input type="checkbox">`

Purpose

Select multiple options.

Example

☒ HTML

☒ CSS

☐ JavaScript

10. Radio Button

`<input type="radio">`

Purpose

Select only one option from a group.

Example

☐ Male

☒ Female

★ Interview Question

Difference Between Checkbox and Radio Button

Checkbox

Radio

Multiple selections Single selection in a group

Independent choices Mutually exclusive choices (when they share the same name)

placeholder

Example

```
<input  
type="text"  
placeholder="Enter your name">
```

Purpose

Displays a hint inside the input.

value

Example

```
<input  
type="text"  
value="Ashvi">
```

Purpose

Provides the initial value of the input.

required

Example

```
<input  
type="email"  
required>
```

Purpose

The browser prevents form submission until the field is filled.

readonly

```
<input  
value="India"  
readonly>
```

Purpose

The user can see and select the value but cannot edit it.

disabled

```
<input  
disabled>
```

Purpose

The field cannot be edited or focused, and it is generally not submitted with the form.

Interview Question

Difference Between readonly and disabled

readonly

Cannot edit

**Can receive focus and usually
submitted**

disabled

Cannot edit

**Cannot receive focus and generally not
submitted**

Real Interview

Interviewer:

What is the purpose of the action attribute?

You:

It specifies the destination where the form data is sent after the form is submitted.

Interviewer:

Difference between GET and POST?

You:

GET sends data in the URL and is suitable for searches or non-sensitive requests. POST sends data in the request body and is commonly used for login, registration, and other forms that handle sensitive information.

PART 10 – Advanced Form Elements & Interview Questions

1. <textarea>

Interview Question

What is <textarea>?

Professional Answer

The <textarea> element is used to accept multi-line text input from the user.

Example:

```
<label for="message">Message</label>
```

```
<textarea id="message"></textarea>
```

Output

Difference between <input> and <textarea>

<input>

<textarea>

Single-line input

Multi-line input

Used for name, email **Used for address, feedback, comments**

Interview Trick

Can `<textarea>` have a closing tag?

Answer

✔ Yes.

Example

```
<textarea>Hello</textarea>
```

Unlike `<input>`, `<textarea>` is not a void element.

2. rows and cols

Example

```
<textarea rows="5" cols="30"></textarea>
```

Meaning

- rows → Height (number of text lines)
 - cols → Width (number of characters)
-

3. `<select>`

Interview Question

What is `<select>`?

Answer

The `<select>` element creates a drop-down list.

Example

```
<select>  
  <option>HTML</option>  
  <option>CSS</option>  
  <option>JavaScript</option>  
</select>
```

Output

▼ HTML

Parent-Child Relationship

select

|

|— option

|— option

|— option

Parent

<select>

Children

<option>

4. <option>

Question

What is <option>?

Answer

Represents an item inside a dropdown.

Example

<option>India</option>

Interview Trick

Can <option> exist without <select>?

✗ No.

Normally it should be inside a <select> element.

5. selected

Example

<option selected>India</option>

Purpose

Makes that option selected by default.

6. value

Example

```
<option value="IN">India</option>
```

Interview Question

Difference between

```
<option value="IN">India</option>
```

Answer

Visible to the user

India

Submitted value

IN

7. <optgroup>

Many freshers don't know this.

Example

```
<select>
```

```
<optgroup label="Frontend">
```

```
<option>HTML</option>
```

```
<option>CSS</option>
```

```
</optgroup>
```

```
<optgroup label="Backend">
```

```
<option>Java</option>
```

<option>Python</option>

</optgroup>

</select>

Purpose

Groups related options inside a dropdown.

8. <button>

Question

What is <button>?

Answer

The <button> element creates a clickable button.

Example

<button>Submit</button>

Interview Question

Difference between

<input type="submit">

and

<button>Submit</button>

<input type="submit">

- **Specifically creates a submit button.**
- **Cannot contain HTML elements inside it.**

<button>

- **More flexible.**
- **Can contain text, icons, images, or other inline HTML.**

Example

```
<button>
  
  Send
</button>
```

9. Button Types

Submit

```
<button type="submit">
Submit
</button>
```

Purpose

Submits the form.

Reset

```
<button type="reset">
Reset
</button>
```

Purpose

Resets the form fields to their initial values.

Button

```
<button type="button">
Click
</button>
```

Purpose

Creates a regular button that does nothing by itself. It is commonly used with JavaScript.

★ Interview Question

Default button type inside a form?

Answer

submit

If you don't specify a type, a <button> inside a form acts as a submit button.

10. name

Interviewers love asking this.

Example

```
<input  
name="username">
```

Question

What is name?

Professional Answer

The name attribute identifies the field when the form is submitted.

Without name, the input's value is generally not included in the submitted form data.

Difference between id and name

id

name

Unique identifier in the page

Field name used when submitting data

Used with CSS, JavaScript, and <label for=""> Used to identify submitted form values

11. maxlength

Example

```
<input  
maxlength="10">
```

Purpose

Limits the number of characters the user can type.

12. minlength

Example

```
<input  
minlength="5">
```

Purpose

Requires at least 5 characters before the form can be submitted.

13. min

Example

```
<input  
type="number"  
min="18">
```

User cannot enter a value smaller than 18 (subject to browser validation).

14. max

Example

```
<input  
type="number"  
max="100">
```

Maximum value

100

15. step

Example

```
<input  
type="number"  
step="5">
```

Allowed values

5

10

15

20

16. autocomplete

Example

```
<input  
autocomplete="on">
```

Purpose

Allows the browser to suggest previously entered values.

17. autofocus

Example

```
<input  
autofocus>
```

Purpose

Automatically places the cursor in that field when the page loads.

18. pattern

Very important.

Example

```
<input  
pattern="[A-Za-z]+">
```

Purpose

Defines a regular expression that the entered value must match.

Example above

Only letters are allowed.

HTML Validation

Interview Question

Does HTML provide validation?

Answer

Yes.

Examples

- **required**
- **email**
- **minlength**
- **maxlength**
- **pattern**
- **min**
- **max**

These allow browsers to perform basic client-side validation.

Common Interview Questions

1. Difference between id and class?

id	class
-----------	--------------

Unique on a page	Can be shared by multiple elements
-------------------------	---

2. Difference between id and name?

id	name
-----------	-------------

Identifies the element in the page	Identifies the field in submitted form data
---	--

3. Difference between placeholder and value?

placeholder value

Hint text Actual initial value

Example

<input

placeholder="Enter Name">

Shows a hint.

Example

<input

value="Ashvi">

Shows "Ashvi" already filled in.

4. Difference between disabled and readonly?

readonly

disabled

Can focus

Cannot focus

Included in form submission Usually not submitted

5. Which attribute connects <label> and <input>?

Answer

for

and

id

The label's for value must match the input's id.

★ Real Interview Round

Interviewer

Why do we use the name attribute?

You

Because the name attribute identifies the form field when the form is submitted.

Without it, the field's value is generally not sent as part of the submitted form data.

Interviewer

Difference between id and name?

You

The id uniquely identifies an element within the page and is used by CSS, JavaScript, and labels. The name attribute identifies the data when the form is submitted to the server.

Interviewer

Difference between placeholder and value?

You

A placeholder provides a hint to the user. A value sets the initial content of the input field.

Super Tricky Questions

Q1. Can two inputs have the same id?

 No.

An id should be unique within a page.

Q2. Can two inputs have the same name?

 Yes.

For example, radio buttons in the same group must share the same name so only one can be selected.

Example:

```
<input type="radio" name="gender" value="Male">
```

```
<input type="radio" name="gender" value="Female">
```

Q3. Can a form contain another form?

 No.

HTML does not allow nested `<form>` elements.

Q4. Can an input exist without a label?

 **Yes.**

But using a <label> is recommended because it improves usability and accessibility.

PART 11 – Semantic HTML (Interview Masterclass)

This topic is asked in almost every fresher interview.

First Question

What is Semantic HTML?

Beginner Answer

Semantic HTML means using HTML tags according to their meaning.

Correct.

Professional Interview Answer

Semantic HTML uses tags that clearly describe the purpose and meaning of the content. These tags make the webpage easier to understand for developers, browsers, search engines, and assistive technologies.

Interview Question

Why do we need Semantic HTML?

Imagine this code:

<div>

<div>

<div>

<div>

</div>

`</div>`

`</div>`

`</div>`

Can you understand what this webpage contains?

✗ No.

Now look:

`<header>`

`</header>`

`<nav>`

`</nav>`

`<main>`

`</main>`

`<footer>`

`</footer>`

Immediately you understand the structure.

That's why Semantic HTML exists.

Benefits of Semantic HTML

Interviewers love this question.

1. Better Readability

Developers can understand the code quickly.

2. Better SEO

Search engines better understand the page structure.

3. Better Accessibility

Screen readers can identify different sections of the page more easily.

4. Easier Maintenance

Future developers can update the code more easily.

Semantic vs Non-Semantic

Non-Semantic

<div>

<div>

<div>

</div>

</div>

</div>

No meaning.

Semantic

<header>

<nav>

</nav>

</header>

<main>

<section>

</section>

</main>

<footer>

</footer>

Each tag has a purpose.

Interview Question

What is the difference between Semantic and Non-Semantic tags?

Answer

Semantic	Non-Semantic
Has meaning	No specific meaning
Describes content	Generic container
Better for SEO	Less descriptive
Better for accessibility	Doesn't describe structure

Examples

Semantic

header

nav

main

section

article

footer

Non-semantic

div

span

<header>

Interview Question

What is <header>?

Answer

The <header> element represents introductory content for a webpage or a section.

It usually contains:

- **Website logo**
- **Website title**
- **Navigation**
- **Search bar**

Example

<header>

<h1>Amazon</h1>

<nav>

...</nav>

</header>

Interview Trick

Can a page have multiple <header> elements?

 **Yes.**

A webpage can have one main header, and individual sections or articles can also have their own headers.

<nav>

Question

What is <nav>?

Answer

Represents a section containing navigation links.

Example

<nav>

Home

About

Contact

</nav>

Purpose

Helps users move between pages or sections.

Interview Question

Should every group of links be inside <nav>?

Answer

No.

Use <nav> only for major navigation links, not for every collection of links.

<main>

Question

Purpose?

Answer

Contains the main content of the webpage.

Everything unique to that page belongs here.

Example

<main>

<h1>HTML Tutorial</h1>

<p>Learning HTML...</p>

</main>

Interview Question

Can a webpage have two <main> elements?

✗ No.

A page should have only one <main> element.

<section>

Question

What is <section>?

Answer

Represents a thematic group of related content.

Example

<section>

<h2>About Us</h2>

<p>Company Information</p>

</section>

Purpose

Groups related content together.

Interview Trick

Should every <section> have a heading?

Best practice: Yes, whenever possible. A section usually has a heading (<h1>–<h6>) describing its topic.

<article>

Very important.

Question

What is <article>?

Professional Answer

Represents independent, self-contained content that could stand on its own.

Examples

- News article
- Blog post
- Forum post
- Product review

Example

`<article>`

`<h2>HTML Interview Tips</h2>`

`<p>...</p>`

`</article>`

Interview Question

Difference between `<section>` and `<article>`?

Section

Article

Groups related content

Self-contained content

May not make sense alone **Can stand alone**

Example

Website

Technology

Inside

AI

Web Development

Cyber Security

Each individual topic can be an <article>.

The whole "Technology" area can be a <section>.

<aside>

Question

Purpose?

Answer

Represents content related to the main content but not part of it.

Examples

- **Advertisements**
 - **Sidebar**
 - **Related articles**
 - **Author information**
-

Example

<aside>

Related Articles

</aside>

<footer>

Question

Purpose?

Answer

Represents the footer of a webpage or a section.

Usually contains

- **Copyright**
 - **Contact information**
 - **Privacy Policy**
 - **Terms and Conditions**
 - **Social media links**
-

Example

<footer>

© 2026 Company

</footer>

Interview Trick

Can there be multiple <footer> elements?

 Yes.

Like <header>, each section or article can have its own footer, and the page can also have a main footer.

Typical Webpage Structure

<body>

<header>

<nav>

</nav>

</header>

<main>

<section>

<article>

</article>

</section>

<aside>

</aside>

</main>

<footer>

</footer>

</body>

Parent-Child Relationship

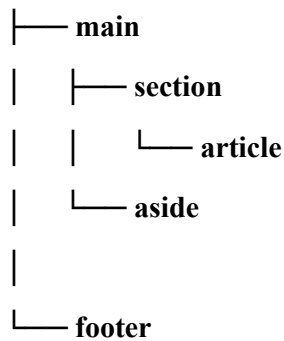
body

|

| — **header**

| | — **nav**

|



Real Interview

Interviewer

What is Semantic HTML?

You

Semantic HTML means using HTML elements according to their meaning so that developers, browsers, search engines, and assistive technologies can better understand the webpage structure.

Interviewer

Difference between `<section>` and `<article>`?

You

A `<section>` groups related content under a common theme, while an `<article>` represents independent content that can stand on its own, such as a blog post or news article.

Interviewer

Can a webpage have multiple `<header>` elements?

You

Yes. A webpage can have one main header, and sections or articles can also have their own headers.

Interviewer

Can a webpage have two `<main>` elements?

You

No. A webpage should contain only one `<main>` element.

★ Super Tricky Questions

1. Is `<div>` a semantic tag?

✗ No.

It is a non-semantic element because it doesn't describe its content.

2. Is `<span>` semantic?

✗ No.

It is also non-semantic.

3. Which semantic tag usually contains navigation links?

Answer

`<nav>`

4. Which semantic tag contains the primary content of the page?

Answer

`<main>`

5. Which semantic tag contains independent content?

Answer

`<article>`

"Why should we use semantic HTML instead of only `<div>` tags?"

A strong answer is:

"Semantic HTML makes the code more meaningful and easier to understand. It improves readability, accessibility for screen readers, and helps search engines understand the page structure better. While `<div>` is useful as a generic container, semantic elements like `<header>`, `<main>`, and `<article>` clearly describe the purpose of each part of the page."

PART 12 – `<div>` vs `<span>` + Block vs Inline Elements

First Question

What is <div>?

Beginner Answer

<div> is a container.

Correct.

Professional Interview Answer

The <div> element is a non-semantic block-level container used to group HTML elements for styling, layout, or scripting.

Let's understand every word.

What is a Container?

Imagine your room.

You have

- **Books**
- **Laptop**
- **Clothes**
- **Charger**

If everything is scattered, your room is messy.

Now you place everything inside a box.

That box is a container.

Similarly,

<div>

groups HTML elements together.

Example

<div>

<h1>Student Details</h1>

<p>Name : Ashvi</p>

<button>Submit</button>

</div>

Everything belongs to one group.

Parent-Child Relationship

div

|

|— **h1**

|— **p**

|— **button**

Parent

<div>

Children

<h1>

<p>

<button>

Why do we use <div>?

Interview Question

Purpose of <div>?

Professional Answer

<div> is mainly used

- To group related elements
 - To apply CSS
 - To create layouts
 - To organize code
 - To target elements with JavaScript
-

Example

```
<div class="card">
```

```
<h2>Employee</h2>
```

```
<p>Developer</p>
```

```
</div>
```

CSS

```
.card{
```

```
background:lightblue;
```

```
padding:20px;
```

```
}
```

The CSS affects the entire group.

Interview Trick

Can <div> contain another <div>?

Answer

 Yes.

Example

<div>

<div>

<div>

</div>

</div>

</div>

This is called nested divs.

Can <div> contain headings?

Yes.

Can it contain paragraphs?

Yes.

Can it contain forms?

Yes.

Can it contain tables?

Yes.

Can it contain images?

Yes.

Can it contain videos?

Yes.

Professional Answer

<div> can contain almost any HTML element.

Is <div> Semantic?

Answer

✗ No.

It doesn't describe the content.

It is simply a generic container.

What is ?

Interview Question

Professional Answer

** is a non-semantic inline container used to group or style a small portion of text or inline content.**

Example

<p>

I love

HTML

</p>

Suppose CSS

```
span{
```

```
color:red;
```

```
}
```

Only

HTML

becomes red.

Why use ?

Purpose

- **Style a word**
 - **Style a sentence**
 - **Apply CSS**
 - **Apply JavaScript**
 - **Group inline content**
-

Interview Question

Can contain text?

Yes.

Can it contain images?

Yes.

Can it contain another ?

Yes.

Can it contain a <div>?

✗ No (best practice).

Because `<div>` is a block-level element, while `<span>` is an inline element. HTML5 parsing rules are more flexible than older versions, but in interviews and good coding practice, you should treat `<span>` as containing only phrasing (inline) content.

Difference Between `<div>` and `<span>`

<code><div></code>	<code></code>
Block-level	Inline
Takes full available width	Takes only the width it needs
Used for larger sections	Used for small text or inline content
Can contain most HTML elements	Intended for inline/phrasing content

Interview Question

Which is block-level?

Answer

`<div>`

Which is inline?

Answer

`<span>`

What is a Block-Level Element?

This is one of the most common interview questions.

Professional Answer

A block-level element starts on a new line and, by default, stretches to fill the available width of its parent.

Example

`<h1>Heading</h1>`

<p>Paragraph</p>

<div>Container</div>

Output

Heading

Paragraph

Container

Every element starts on a new line.

Characteristics of Block Elements

- ✓ Starts on a new line**
- ✓ Takes full available width by default**
- ✓ Can have width and height**
- ✓ Can contain block and inline elements (depending on the specific HTML content model)**

Common Block Elements

<div>

<p>

<h1>

<h2>

<h3>

<table>

<form>

<header>

<footer>

<section>

article

nav

main

What is an Inline Element?

Professional Answer

An inline element does not start on a new line and only takes up as much width as its content requires.

Example

I love

HTML

and

CSS.

Output

I love HTML and CSS.

Everything appears on the same line.

Characteristics of Inline Elements

- ☒ **Doesn't start a new line**
 - ☒ **Only occupies required width**
 - ☒ **Usually used inside text**
-

Common Inline Elements

<a>

`<i>`

`<label>`

`<code>`

`<abbr>`

★ Interview Question

Difference Between Block and Inline

Block	Inline
Starts on a new line	Continues on the same line
Takes full available width	Takes only required width
Used for larger page structure	Used for text and inline content

Real Interview

Interviewer

What is `<div>`?

You

`<div>` is a non-semantic block-level container used to group HTML elements for layout, styling, or scripting.

Interviewer

What is `<span>`?

You

`<span>` is a non-semantic inline container used to group or style a small portion of text or other inline content.

Interviewer

Difference between `<div>` and `<span>`?

You

`<div>` is block-level and is used for larger sections of a page, while `<span>` is inline and is used to style or group small pieces of text or inline content.

Interviewer

Give five examples of block-level elements.

You

- `<div>`
 - `<p>`
 - `<h1>`
 - `<form>`
 - `<table>`
-

Interviewer

Give five examples of inline elements.

You

- `<span>`
 - `<a>`
 - `<img>`
 - `<strong>`
 - `<em>`
-

Super Tricky Questions

1. Is `<img>` block or inline?

 By default, `<img>` is an inline element (more precisely, an inline replaced element).

2. Is `<a>` block or inline?

 By default, inline.

3. Can CSS change block to inline?

✓ Yes.

Example

```
div{  
display:inline;  
}
```

4. Can CSS change inline to block?

✓ Yes.

Example

```
span{  
display:block;  
}
```

5. Is <div> semantic?

✗ No.

6. Is semantic?

✗ No.

★ Coding Example

```
<div class="profile">
```

```
<h2>Ashvi</h2>
```

```
<p>
```

Learning

```
<span>HTML</span>
```

and

`<span>CSS</span>`

`</p>`

`</div>`

Here:

- `<div>` groups the whole profile.
- `<span>` highlights individual words.
- "Why not use only `<div>` everywhere?"
- A strong answer is:
- "While `<div>` is useful as a generic container, using only `<div>` makes the HTML less meaningful. Semantic elements like `<header>`, `<main>`, `<section>`, and `<article>` improve readability, accessibility, and SEO. I use `<div>` when I need a generic container and semantic elements when the content has a specific purpose."

PART 13 – HTML Entities, `<iframe>`, `<audio>`, and `<video>`

Topic 1 – HTML Entities

Interview Question

What are HTML Entities?

Professional Answer

HTML entities are special codes used to display reserved characters or symbols that cannot be typed directly in HTML.

Why do we need HTML Entities?

Suppose you write:

`<p><h1></p>`

The browser thinks `<h1>` is an HTML tag.

But what if you want to display this text on the webpage?

`<h1>`

Instead of creating a heading?

We use HTML entities.

Most Important HTML Entities

Less Than (<)

<

Output

 $\angle$

Greater Than ($>$)

>

Output

 $\succ$

Ampersand (&)

&

Output

&

Non-breaking Space

Purpose

Creates a space that the browser won't collapse.

Example

Ashvi HTML

Output

Ashvi HTML

Copyright Symbol

©

Output

©

Registered Trademark

®

Output

®

Trademark

™

Output

™

Rupee Symbol

₹

Output

₹

Interview Question

How do you display <h1> as text?

Answer

<h1>

Output

<h1>

Topic 2 – <iframe>

One of the most asked practical questions.

What is <iframe>?

Professional Answer

The `<iframe>` element is used to embed another webpage or external content inside the current webpage.

Example

```
<iframe  
src="https://example.com">  
</iframe>
```

Common Uses

- YouTube videos
 - Google Maps
 - Another website (if allowed)
 - PDF documents
-

Example – YouTube

```
<iframe  
width="560"  
height="315"  
src="https://www.youtube.com/embed/VIDEO_ID">  
</iframe>
```

Example – Google Maps

```
<iframe  
src="Google Maps Embed URL">  
</iframe>
```

Important Attributes

src

Specifies the URL to display.

width

Sets the width.

height

Sets the height.

title

Provides a description for accessibility.

Example

<iframe

title="Company Location"

src="...">

</iframe>

Interview Trick

Can every website be opened inside an iframe?

✗ No.

Many websites block embedding using security settings (such as X-Frame-Options or Content Security Policy).

Topic 3 – <audio>

What is <audio>?

Professional Answer

The <audio> element is used to embed and play audio files.

Example

<audio controls>

```
<source  
src="song.mp3"  
type="audio/mpeg">  
  
</audio>
```

Why <source>?

Because one audio element can provide multiple file formats.

Example

```
<audio controls>
```

```
<source  
src="song.mp3"  
type="audio/mpeg">
```

```
<source  
src="song.ogg"  
type="audio/ogg">
```

```
</audio>
```

The browser chooses the first supported format.

Important Attributes

controls

Shows play/pause buttons.

autoplay

Starts automatically.

⚠ Many browsers restrict autoplay unless the audio is muted or the user has interacted with the page.

loop

Repeats continuously.

muted

Starts with no sound.

Topic 4 – <video>

What is <video>?

Professional Answer

The <video> element is used to embed and play video files.

Example

<video

width="400"

controls>

<source

src="movie.mp4"

type="video/mp4">

</video>

Important Attributes

controls

Shows

- **Play**
- **Pause**
- **Volume**
- **Full Screen**

autoplay

Starts automatically (subject to browser restrictions).

loop

Repeats the video.

muted

Starts silently.

poster

Shows an image before the video plays.

Example

<video

poster="thumbnail.jpg">

★ Difference Between <audio> and <video>

<audio>

Plays sound

No visual content

Supports audio formats

<video>

Plays video with sound

Displays visual content

Supports video formats

★ Real Interview Questions

Interviewer

What are HTML entities?

You

HTML entities are special codes used to display reserved characters or symbols that cannot be typed directly in HTML.

Interviewer

Why do we use <?

You

To display the < character as text instead of having the browser interpret it as the start of an HTML tag.

Interviewer

What is an iframe?

You

An iframe is an HTML element used to embed another webpage or external content inside the current webpage.

Interviewer

Can every website be opened inside an iframe?

You

No. Many websites prevent embedding for security reasons using HTTP headers such as X-Frame-Options or Content Security Policy.

Interviewer

Difference between audio and video?

You

The <audio> element is used to play audio files, while the <video> element is used to play video files with visual content.

ART 14 – Miscellaneous HTML5 Tags (Interview Preparation)

1. <details>

Interview Question

What is the <details> tag?

Professional Answer

The <details> element creates a collapsible section that users can expand or collapse.

Example

<details>

<summary>What is HTML?</summary>

<p>HTML stands for HyperText Markup Language.</p>

</details>

Output

► What is HTML?

(click)

▼ What is HTML?

HTML stands for HyperText Markup Language.

Why do we use it?

- FAQ sections
 - Show/Hide Answers
 - Instructions
 - Notes
-

2. <summary>

Question

What is <summary>?

Answer

It provides the heading for a <details> element.

Without <summary>, the browser shows a generic label.

Parent Relationship

details



Interview Trick

Can <summary> exist without <details>?

✗ It is intended to be used as the first child of a <details> element.

3. <progress>

Very common in projects.

Question

What is <progress>?

Answer

Displays the progress of a task.

Example

<progress value="70" max="100"></progress>

Output



70%

Interview Question

Difference between value and max

value max

Current progress Maximum progress

4. <meter>

Many students confuse this with <progress>.

Example

```
<meter value="0.8">80%</meter>
```

Purpose

Represents a measurement within a known range.

Examples

- Battery Level
 - Disk Usage
 - Fuel Level
 - Exam Score
-

Difference Between <progress> and <meter>

<progress> <meter>

Progress of a task Measurement within a range

Loading file Battery percentage

Example

Downloading a file



60%

Use

```
<progress>
```

Battery

 80%

Use

<meter>

5. <abbr>

Question

Purpose?

Answer

Represents an abbreviation.

Example

<abbr title="HyperText Markup Language">

HTML

</abbr>

When the user hovers over "HTML", many browsers display:

HyperText Markup Language

6. <code>

Question

Purpose?

Answer

Displays code snippets.

Example

<code>

console.log("Hello");

</code>

Output

console.log("Hello");

Usually shown in a monospace font.

7. `<kbd>`

Very interesting.

Question

Purpose?

Answer

Represents keyboard input.

Example

Press

`<kbd>Ctrl</kbd>`

+

`<kbd>C</kbd>`

to copy.

Output

Ctrl + C

8. `<mark>`

You already learned this as a formatting tag.

Purpose

Highlights text.

Example

`<p>`

Learning

<mark>HTML</mark>

today.

</p>

9. <time>

Question

Purpose?

Answer

Represents dates or times.

Example

<time datetime="2026-07-29">

July 29, 2026

</time>

Helps browsers and search engines understand that the content is a date.

10. <address>

Question

Purpose?

Answer

Represents contact information for the author, organization, or owner of the page.

Example

<address>

Email:

abc@gmail.com

</address>

11. <blockquote>

Question

Purpose?

Answer

Represents a quotation from another source.

Example

<blockquote>

Success is not final.

</blockquote>

12. <cite>

Question

Purpose?

Answer

Represents the title of a creative work.

Example

<cite>

Harry Potter

</cite>

13. `<small>`

Question

Purpose?

Answer

Displays side comments or fine print.

Example

`<small>`

Terms and Conditions Apply

`</small>`

14. `<sup>` and `<sub>`

Superscript

`x<sup>2</sup>`

Output

x^2

Subscript

`H<sub>2</sub>O`

Output

H₂O

★ Real Interview Questions

Interviewer

Difference between `<progress>` and `<meter>`?

You

`<progress>` shows how much of a task has been completed, while `<meter>` represents a value within a known range, such as battery level or disk usage.

Interviewer

What is <abbr>?

You

It represents an abbreviation or acronym and can provide the full form using the title attribute.

Interviewer

What is <kbd>?

You

It represents keyboard input, such as shortcut keys like Ctrl + C.

Interviewer

Why do we use <time>?

You

To represent dates and times in a machine-readable way, helping browsers, assistive technologies, and search engines understand the content.

Super Tricky Questions

1. Is <details> semantic?

☒ **Yes.**

2. Is <summary> semantic?

☒ **Yes.**

3. Which tag is used for keyboard shortcuts?

Answer

<kbd>

4. Which tag displays program code?

Answer

<code>

5. Which tag represents an abbreviation?

Answer

<abbr>
